Unit 3 - Calculus
3.5 Optimization

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Outline

1 Gradient Descent

Gradient Descent

Gradient Descent

Optimization:

Optimization is the process of finding the minimum/maximum value of a function $f(x)$ over some domain D:

$$\min_{x \in D} f(x) \qquad \max_{x \in D} f(x)$$

- f is called the **objective function**
- D is called the search space
- A value of x which minimizes/maximizes f is called a minimum/maximum (more generally, extremum)

Applications of Optimization

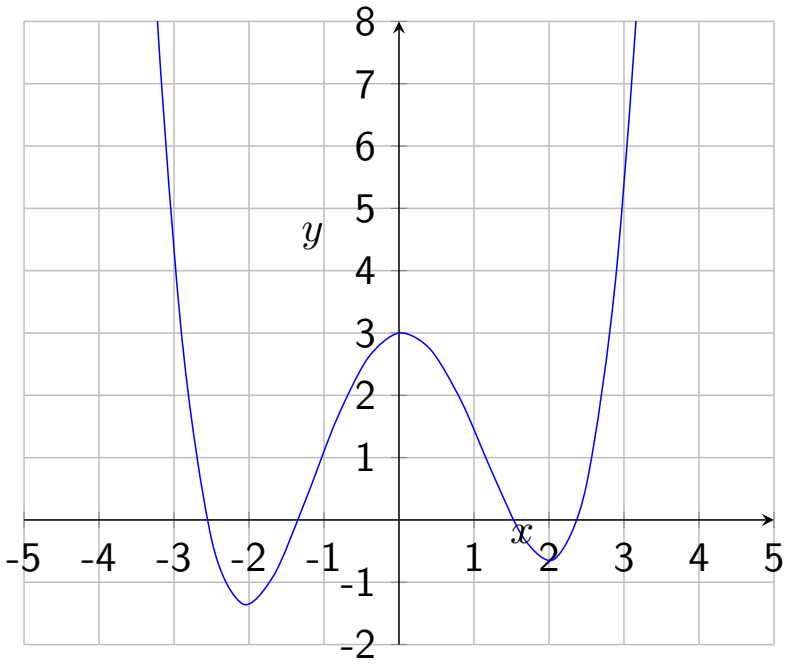
Applications of optimization:

- Engineering: used in computer-aided design (e.g., finding the ideal plane wing to minimize drag but maximize lift)
- Physics: used to fit models of physical systems to observed data (my research!)
- Machine learning: used to train models (e.g., minimizing the loss function in neural networks)
- Operations research: used to solve problems in logistics, resource allocation, scheduling, and other areas of business and industry
- Finance: used to optimize investment portfolios and manage risk
- Game theory: used to model and analyze strategic interactions between agents

Global vs. Local Minima

Global vs. local:

- A **global minimum** is a value x^* such that $f(x^*) \leq f(x)$ for all $x \in D$
- A **local minimum** is a value x^* such that $f(x^*) \leq f(x)$ for all x in a neighborhood of x^* (meets the calculus criteria for extremum, but isn't an extremal value)



Optimization

Gradient descent (GD):

- Gradient descent is an optimization method which uses the gradient/derivative of a function to find its minimum
- Given a starting point (x_0), it iteratively generates points by traveling along the direction of the function's steepest descent, i.e., the negative of the gradient
- The learning rate α (usually less than 1) controls the speed of descent

Algorithm Gradient descent

1: Set: learning rate α, number of iterations N
2: Set: x_0
3: **for** $i = 0$ to $N - 1$ **do**
4: $\quad x_{i+1} = x_i - \alpha f'(x_i)$
5: **end for**
6: Return: x_N

Optimization

Advantages of gradient descent:

- Simple implementation
- Step size scales with size of gradient for fast convergence

Problems with gradient descent:

- Get's easily stuck in local minima
- Can oscillate between solutions
- Gradients can explode/vanish
- Requires a formula for the derivative or a numerical approximation